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REDUCED ORDER MODELS OF LOW MACH NUMBER ISOTHERMAL FLOWS IN MICROCHANNELS

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ABSTRACT

We present reduced order models of unsteady low Mach number isothermal ideal gas flows in two-dimensional rectangular microchannels subject to first order slip boundary conditions. The Navier-Stokes equations are simplified using Low Mach Number expansions of the pressure and velocity fields. This approximation allows decoupling the density from spatial pressure variations, thus simplifying the momentum equation. The resulting diffusion equation and the subsequent pressure-flow-rate relationship enables modeling the flow using analog circuit components. The accuracy of the proposed models is investigated for steady and unsteady flows in a two-dimensional channel for different values of Reynolds and Knudsen numbers.

NOMENCLATURE

A Channel cross sectional area.
 h Height of rectangular channel cross section.
 L Channel length.
 p Pressure.
 Q Volume flow rate.
 R Ideal gas constant.
 α Channel aspect ration (h/L).
 Re_h Reynolds number based on channel's height.
 St_h Strouhal number based on channel's height.
 Ma Mach number.
 Kn Knudsen number.

t Time.
 u Velocity component in flow direction
 x Coordinate in flow direction.
 y Coordinate along channel height.
 $\mathbf{u}$ Velocity vector.
 λ Mean free path.
 μ Dynamic viscosity.
 ρ Density.
 γ Ratio of constant pressure to constant volume specific heat.
 σ_v Accommodation coefficient.

INTRODUCTION

Early interest in micro flows originated in attempts to model the airways in the respiratory system. More recently, with the advent of Micro-Electro-Mechanical Systems (MEMS), modeling gas micro-flows became important in many applications such as micro-compressors, micro-valves, micro-filters, micro-combustors [1], micro-turbines [2], microthrusters, and micro-nozzles. The dynamic role of gas surrounding an oscillating microstructure [3] is also an important part in designing robust devices such as micro RF switches and resonators. The complexity of these devices makes simulation of the detailed physics an expensive and a time consuming task. Alternative design and analysis approaches based on reduced order modeling [4] are attractive due the advantages they offer in terms of speed, cost, and ability to perform analysis and design in an integrated multiple domain "circuit design" environment. System level modeling

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also enables fast sweeping of the design parameters thereby reducing significantly the design space. Development of efficient yet accurate reduced order models for individual components such as microchannels is paramount to the viability of system level design approach.

Microchannels are the most common building blocks of microfluidic devices. They offer transport of small volumes of the working fluid. As opposed to macroscale flows, gaseous microflows are generally laminar with small but significant value of Reynolds number. In some applications, such as micro turbo-machines [5], Reynolds number could reach a few thousands, indicating the increasing role of inertia effects. Unlike the case with macroscale flows, micro gas flows can experience rarefaction (even at small value of Mach number) when the device's dimensions become comparable to the mean free path of the gas, (Knudsen number larger than 0.001). If the departure from continuum is not significant ($Kn \leq 0.1$), the Navier-Stokes equations are still applicable with slip boundary condition [6–8] at a solid wall.

Because of large surface to volume ratio of microdevices, it is reasonable to assume that the flow is isothermal if the channel wall is a good heat conductor such as silicon [7]. Reduced order dynamic modeling of microchannel isothermal gas flows requires capturing the unsteady behavior of the flow accurately. This is particularly important in applications in which precise control of the flow conditions is required. For low inertia isothermal gas flows in large aspect ratio channels ($\alpha \ll 1, Re < 1, ReSt \ll 1$), pressure drop balances the shear force resulting in the well known pressure-flow rate relation [7]

$$Q = \frac{h^3}{12\mu} \left(-\frac{\partial p}{\partial x} \right) \left(1 + 6 \frac{2-\sigma_v}{\sigma_v} Kn_0 \frac{p_0}{p} \right) \quad (1)$$

where the subscript zero refer to conditions at the reference temperature T_0 . The analog circuit model, shown in Fig. 1, corresponds to a voltage controlled resistor $R = \frac{12\mu}{h^3} \left(1 + 6 \frac{2-\sigma_v}{\sigma_v} Kn_0 \frac{p_0}{p} \right)^{-1}$ per unit length, where the volume flow rate is taken as the through variable (current) and the pressure is taken as the across variable (voltage).

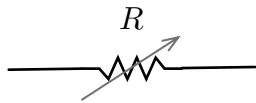


Figure 1. Inertia-free isothermal model.

Most models available in the literature account for **steady** isothermal gas flow through parallel channels [9–15]. Those

range from slightly compressible [15] to compressible subsonic flows [13, 14, 16]. Among those, the authors in [16] present an model based on Burnett momentum equation of a steady isothermal, compressible subsonic gas flow in a micro channel with slowly varying cross-section whereas the authors in [17] relax the isothermal assumption but deal with a parallel flow. None of the solutions in the aforementioned studies is assimilated in a circuit model but most importantly none of them deals with unsteady flows.

A system-level circuit model for an **unsteady** incompressible isothermal flow in channels of slightly varying cross-section is presented in [18]. This work can be viewed as an extension of [18] where the objective is to present a novel dynamic analog circuit model for isothermal low mach number gas flow in a two dimensional long channels where inertia effects are substantial ($\alpha \ll 1, Re_h St_h \geq 0, \alpha Re_h \geq 0$). The derived models are compared to the inertia-free model of Eqn. (1) and numerical results from CFD simulations using Ansys (Fluent).

GOVERNING EQUATIONS IN DIMENSIONLESS FORM FOR AN ISOTHERMAL GAS

The conservation of mass, momentum for a single species non-reacting Newtonian fluid are

$$\begin{aligned} \frac{D\rho}{Dt} + \rho \nabla \cdot \vec{u} &= 0 \\ \rho \left(\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} \right) &= -\nabla p + \mu (\Delta \vec{u} + \frac{1}{3} \nabla (\nabla \cdot \vec{u})) \end{aligned} \quad (2)$$

In what follows, we denote dimensional variables by a tilde accent. Assume the flow direction along the channel in the x -direction. We non-dimentionalize according the following scales

1. Length scales: $x \sim L, y \sim h_0$
2. Velocity $u \sim u_0, v \sim v_0 = \beta u_0$ (The continuity equation may be used to come up with appropriate scaling of the y -component of the velocity)
3. Pressure and density : $p \sim p_0, \rho \sim \rho_0$.
4. Time scale: $t \sim 1/f$

We also assume isothermal ideal gas flow where dimensional pressure and temperature are related by ideal gas law $\tilde{p} = \tilde{\rho} R \tilde{T}_0$, $\tilde{p}_0 = \tilde{\rho}_0 R \tilde{T}_0$ so that $p = \rho$ (since $\tilde{p}/\tilde{p}_0 = \tilde{\rho}/\tilde{\rho}_0$) and $\frac{\tilde{p}_0/\tilde{\rho}_0}{\tilde{u}_0^2} = \frac{1}{\gamma Ma^2}$. We then obtain the dimensionless form of the mass, x and y mo-

mentum after neglecting terms of order $\alpha^2, \alpha\beta$ and higher

$$\begin{aligned} St_h \frac{\partial p}{\partial t} + \alpha \frac{\partial(pu)}{\partial x} + \beta \frac{\partial(pv)}{\partial y} &= 0 \\ Re_h p \left[St_h \frac{\partial u}{\partial t} + \alpha u \frac{\partial u}{\partial x} + \beta v \frac{\partial u}{\partial y} \right] \\ &\simeq -\alpha \frac{Re_h}{\gamma Ma^2} \frac{\partial p}{\partial x} + \frac{\partial^2 u}{\partial y^2} \\ \beta Re_h p \left[St_h \frac{\partial v}{\partial t} \right] \\ &\simeq -\frac{Re_h}{\gamma Ma^2} \frac{\partial p}{\partial y} + \beta \frac{\partial^2 v}{\partial y^2} + \frac{1}{3} \frac{\partial}{\partial y} \left(\alpha \frac{\partial u}{\partial x} + \beta \frac{\partial v}{\partial y} \right), \end{aligned}$$

subject to the boundary conditions

$$\text{at } y = 0: \frac{\partial u}{\partial y} = 0 \text{ and } v = 0,$$

and to the first order slip boundary condition at the wall

$$u|_{y=h(x)/2} = -\frac{2-\sigma_v}{\sigma_v} \frac{Kn}{p} \frac{\partial u}{\partial y} \Big|_{y=h(x)/2}, \text{ and } v|_{y=h(x)/2} = 0.$$

We have used the following dimensionless variables:

Reynolds number, measures inertia to viscosity: $Re_h = \frac{\rho_0 u_0 h_0}{\mu}$

Strouhal's number, measures unsteady (oscillating) to advection components of inertia: $St_h = \frac{f h_0}{u_0}$

Mach number, measures the ratio of the characteristic flow speed to the local speed of sound: $Ma = \frac{u_0}{\sqrt{\gamma RT}}$

Knudsen number is $Kn = \sqrt{\frac{\gamma \pi}{2}} \frac{Ma}{Re_h}$.

LOW MACH NUMBER PARALLEL FLOW

Parallel channel flow

We now seek an expansion in terms of β , that is we seek a solution of the form

$$\begin{aligned} p &= p^0 + O(\beta^1) \\ u &= u^0 + O(\beta^1) \\ v &= v^0 + O(\beta^1) \end{aligned} \quad (3)$$

Putting this expansion in our equations, we get to $O(\beta^0)$, the

equations for mass, x and y- momentum:

$$\begin{aligned} St_h \frac{\partial p^0}{\partial t} + \alpha \frac{\partial}{\partial x} (p^0 u^0) &\simeq 0 \\ Re_h p^0 \left[St_h \frac{\partial u^0}{\partial t} + \alpha u^0 \frac{\partial u^0}{\partial x} \right] &\simeq -\alpha \frac{Re_h}{\gamma Ma^2} \frac{\partial p^0}{\partial x} + \frac{\partial^2 u^0}{\partial y^2} \\ -\frac{Re_h}{\gamma Ma^2} \frac{\partial p^0}{\partial y} + \frac{\alpha}{3} \frac{\partial}{\partial y} \left(\frac{\partial u^0}{\partial x} \right) &\simeq 0 \end{aligned}$$

where the boundary condition at the centerline is

$$\frac{\partial u^0}{\partial y} \simeq 0.$$

and the boundary Condition at the wall

$$(p^0 u^0)|_{y=1/2} \simeq -\frac{2-\sigma_v}{\sigma_v} Kn \frac{\partial u^0}{\partial y} \Big|_{y=1/2}$$

Low Mach number expansion

We then expand our solution in terms of $\epsilon = \sqrt{\gamma} Ma$, regarding $Kn = \sqrt{\frac{\pi}{2}} \frac{\sqrt{\gamma} Ma}{Re_h}$ as $> O(\epsilon)$, that is we seek a solution of the system

$$\begin{aligned} \text{mass} \quad St_h \frac{\partial p^0}{\partial t} + \alpha \frac{\partial}{\partial x} (p^0 u^0) &\simeq 0 \\ x - \text{mom} \quad Re_h p^0 \left[St_h \frac{\partial u^0}{\partial t} + \alpha u^0 \frac{\partial u^0}{\partial x} \right] &\simeq -\alpha \sqrt{\frac{\pi}{2}} \frac{1}{\epsilon Kn} \frac{\partial p^0}{\partial x} + \frac{\partial^2 u^0}{\partial y^2} \\ y - \text{mom} \quad \frac{\partial}{\partial y} \left(-\sqrt{\frac{\pi}{2}} \frac{1}{\epsilon Kn} p^0 + \frac{\alpha}{3} \frac{\partial u^0}{\partial x} \right) &\simeq 0 \\ \text{at } y = 0: \quad \frac{\partial u^0}{\partial y} &\simeq 0 \\ \text{at } y = 1/2: (p^0 u^0) &\simeq -\frac{2-\sigma_v}{\sigma_v} Kn \frac{\partial u^0}{\partial y} \end{aligned}$$

in the form

$$\begin{aligned} p^0 &= p^{0,0} + \epsilon p^{0,1} + \epsilon^2 p^{0,2} + O(\epsilon^3) \\ u^0 &= u^{0,0} + \epsilon u^{0,1} + \epsilon^2 u^{0,2} + O(\epsilon^3) \\ v^0 &= v^{0,0} + \epsilon v^{0,1} + \epsilon^2 v^{0,2} + O(\epsilon^3) \end{aligned} \quad (4)$$

We obtain the following system, combining $O(\epsilon^0)$ and $O(\epsilon)$

equations

$$\begin{aligned}
\text{St}_h \frac{\partial p^{0,0}}{\partial t} + \alpha \frac{\partial}{\partial x} (p^{0,0} u^{0,0}) &= 0 \\
\frac{\partial p^{0,0}}{\partial x} &= 0, \\
\frac{\partial p^{0,0}}{\partial y} &= 0 \implies p^{0,0} \equiv p^{0,0}(t) \\
\text{Re}_h p^{0,0} \left[\text{St}_h \frac{\partial u^{0,0}}{\partial t} + \alpha u^{0,0} \frac{\partial u^{0,0}}{\partial x} \right] &= -\alpha \sqrt{\frac{\pi}{2}} \frac{1}{\text{Kn}} \frac{\partial p^{0,1}}{\partial x} + \frac{\partial^2 u^{0,0}}{\partial y^2}
\end{aligned} \tag{5}$$

satisfying the boundary conditions

$$\begin{aligned}
\frac{\partial u^{0,0}}{\partial y} &= 0, \text{ at } y = 0 \\
u^{0,0} p^0 &= -\frac{2-\sigma_v}{\sigma_v} \text{Kn} \frac{\partial u^{0,0}}{\partial y} \text{ at } y = 1/2
\end{aligned}$$

THE REDUCED ORDER MODEL

We simplify the mass equation to

$$-\text{St}_h \frac{dp^{0,0}}{dt} = \alpha p^{0,0} \frac{\partial u^{0,0}}{\partial x}$$

and use it the momentum Eqn.(5) to get

$$\text{Re}_h \text{St}_h \left[p^{0,0} \frac{\partial u^{0,0}}{\partial t} - \frac{dp^{0,0}}{dt} u^{0,0} \right] = -\alpha \sqrt{\frac{\pi}{2}} \frac{1}{\text{Kn}} \frac{\partial p^{0,1}}{\partial x} + \frac{\partial^2 u^{0,0}}{\partial y^2}. \tag{6}$$

Let us use the following integral transform in the space variable y :

$$\hat{u}(x, \beta_m, t) = \int_0^{1/2} \mathcal{K}(\beta_m, y') u^{0,0}(x, y', t) dy'$$

along with the inverse transform

$$u^{0,0}(x, y, t) = \sum_{m=0}^{\infty} \mathcal{K}(\beta_m, y) \hat{u}(x, \beta_m, t),$$

where the kernel solves the eigen-value problem

$$\frac{d^2 \phi}{dy^2} + \beta^2 \phi(y) = 0, \text{ in } 0 \leq y \leq 1/2 \tag{7}$$

subject to the boundary condition

$$\phi'(0) = 0, \text{ and } \frac{2-\sigma_v}{\sigma_v} \text{Kn} \phi'(1/2) + p^0 \phi(1/2) = 0.$$

The solution of Eqn.(7) is

$$\mathcal{K}(\beta_m, y) = \sqrt{2} \left(\frac{\beta_m^2 + \left(\frac{\sigma_v p^0}{(2-\sigma_v) \text{Kn}} \right)^2}{\frac{1}{2} \left[\beta_m^2 + \left(\frac{\sigma_v p^0}{(2-\sigma_v) \text{Kn}} \right)^2 \right] + \frac{\sigma_v p^0}{(2-\sigma_v) \text{Kn}}} \right)^{1/2} \cos \beta_m y, \tag{8}$$

where β_m solves the characteristic equation

$$\beta_m \tan \left(\frac{\beta_m}{2} \right) = \frac{\sigma_v p^0}{(2-\sigma_v) \text{Kn}} \tag{9}$$

Note that the norm of the kernel is

$$\begin{aligned}
N_m &= \int_0^{1/2} \mathcal{K}(\beta_m, y) dy \\
&= \sqrt{2} \left(\frac{\beta_m^2 + \left(\frac{\sigma_v p^0}{(2-\sigma_v) \text{Kn}} \right)^2}{\frac{1}{2} \left[\beta_m^2 + \left(\frac{\sigma_v p^0}{(2-\sigma_v) \text{Kn}} \right)^2 \right] + \frac{\sigma_v p^0}{(2-\sigma_v) \text{Kn}}} \right)^{1/2} \frac{\sin \left(\frac{\beta_m}{2} \right)}{\beta_m}
\end{aligned} \tag{10}$$

Let us now apply this transform to Eqn.(6) (We consider x a parameter here)

$$\text{Re}_h \text{St}_h \left[p^{0,0} \frac{d\hat{u}}{dt} - \frac{dp^{0,0}}{dt} \hat{u} \right] = -\alpha \sqrt{\frac{\pi}{2}} \frac{N_m}{\text{Kn}} \frac{\partial p^{0,1}}{\partial x} - \beta_m^2 \hat{u}.$$

with the initial condition $\hat{u}(t=0) = 0$, which we solve to get

$$\begin{aligned}
\hat{u}(x, \beta_m, t) &= -\sqrt{\frac{\pi}{2}} \frac{\alpha}{\text{St}_h \text{Re}_h \text{Kn}} N_m p^{0,0}(t) e^{-\frac{\beta_m^2}{\text{Re}_h \text{St}_h} \int_0^t \frac{ds}{p^{0,0}(s)}} \times \\
&\int_0^t \frac{\partial p^{0,1}(x, s)}{\partial x} e^{\frac{\beta_m^2}{\text{Re}_h \text{St}_h} \int_0^s \frac{dz}{p^{0,0}(z)}} (p^{0,0}(s))^{-2} ds
\end{aligned}$$

Inverting the transform, we get

$$u(x, y, t) = -\sqrt{\frac{\pi}{2}} \frac{\alpha p^{0,0}(t)}{\text{St}_h \text{Re}_h \text{Kn}} \sum_{m=0}^{\infty} N_m \mathcal{K}(\beta_m, y) e^{-\frac{\beta_m^2}{\text{Re}_h \text{St}_h} \int_0^t \frac{ds}{p^{0,0}(s)}} \times \tag{11}$$

$$\int_0^t \frac{\partial p^{2,0}(x, s)}{\partial x} e^{\frac{\beta_m^2}{\text{Re}_h \text{St}_h} \int_0^s \frac{dz}{p^{0,0}(z)}} (p^{0,0}(s))^{-2} ds,$$

where β_m satisfies the characteristic equation Eqn.(9) and the kernel is given by Eqn.(8).

Upon integrating in y , we get

$$\begin{aligned} Q(x,t) &= 2 \int_0^{1/2} u(x,y,t) dy \\ &= -\sqrt{\frac{\pi}{2}} \frac{\alpha p^{0,0}(t)}{\text{St}_h \text{Re}_h \text{Kn}} \sum_{m=0}^{\infty} N_m^2 e^{-\frac{\beta_m^2}{\text{Re}_h \text{St}_h} \int_0^t \frac{ds}{p^{0,0}(s)}} \times \\ &\quad \int_0^t \frac{\partial p^{2,0}(x,s)}{\partial x} e^{\frac{\beta_m^2}{\text{Re}_h \text{St}_h} \int_0^s \frac{dz}{p^{0,0}(z)}} (p^{0,0}(s))^{-2} ds, \end{aligned}$$

and integrating x , we get

$$\begin{aligned} \bar{Q}(t) &= \int_0^1 Q(x,t) dx \\ &= 2\alpha \sqrt{\frac{\pi}{2}} \frac{p^{0,0}(t)}{\text{St}_h \text{Re}_h \text{Kn}} \sum_{m=0}^{\infty} N_m^2 e^{-\frac{\beta_m^2}{\text{Re}_h \text{St}_h} \int_0^t \frac{ds}{p^{0,0}(s)}} \times \\ &\quad \int_0^t \Delta p(s) e^{\frac{\beta_m^2}{\text{Re}_h \text{St}_h} \int_0^s \frac{dz}{p^{0,0}(z)}} (p^{0,0}(s))^{-2} ds \\ &= \sum_m \bar{Q}_m(t), \end{aligned} \quad (12)$$

where

$$\begin{aligned} \bar{Q}_m(t) &= 2\alpha \sqrt{\frac{\pi}{2}} \frac{p^{0,0}(t)}{\text{St}_h \text{Re}_h \text{Kn}} N_m^2 e^{-\frac{\beta_m^2}{\text{Re}_h \text{St}_h} \int_0^t \frac{ds}{p^{0,0}(s)}} \times \\ &\quad \int_0^t \Delta p(s) e^{\frac{\beta_m^2}{\text{Re}_h \text{St}_h} \int_0^s \frac{dz}{p^{0,0}(z)}} (p^{0,0}(s))^{-2} ds \end{aligned} \quad (13)$$

We see that $\bar{Q}_m(t)$ satisfies:

$$\text{Re}_h \text{St}_h \left[p^{0,0} \frac{d\bar{Q}_m}{dt} - \frac{dp^{0,0}}{dt} \bar{Q}_m \right] = \alpha \sqrt{\frac{\pi}{2}} \frac{N_m}{\text{Kn}} \Delta p^{0,1} - \beta_m^2 \bar{Q}_m.$$

or the circuit model

$$\sqrt{\frac{2}{\pi}} \frac{\text{Kn}}{\alpha N_m} \text{Re}_h \text{St}_h \left[p^{0,0} \frac{d\bar{Q}_m}{dt} + \left(\frac{\beta_m^2}{\text{Re}_h \text{St}_h} - \frac{dp^{0,0}}{dt} \right) \bar{Q}_m \right] = \Delta p \quad (14)$$

The circuit model, based on Eqn.(14), is shown in Fig. 2. It is made up of an infinite number of parallel branches, with each

branch consisting of time dependent inductor L_m in series with a time dependent resistor R_m , where

$$L_m = \sqrt{\frac{2}{\pi}} \frac{\text{Kn}}{\alpha N_m} \text{Re}_h \text{St}_h p^{0,0}(t),$$

and

$$R_m = \sqrt{\frac{2}{\pi}} \frac{\text{Kn}}{\alpha N_m} \text{Re}_h \text{St}_h \left(\frac{\beta_m^2}{\text{Re}_h \text{St}_h} - \frac{dp^{0,0}}{dt} \right).$$

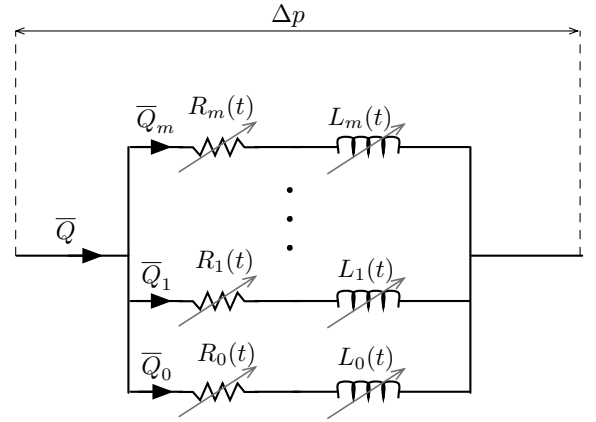


Figure 2. Circuit model.

Note that the steady solution for a constant $p^{0,0}$ and Δp is obtained from Eqn.(12), by taking the limit $t \rightarrow \infty$ as

$$\bar{Q}^{steady} = \sum_m \sqrt{\frac{\pi}{2}} \frac{2\alpha}{\beta_m^2 \text{Kn}} N_m^2 \Delta \quad (15)$$

No slip case

The solution in this case, which corresponds to $(\text{Kn} = O(\epsilon))$ is given by

$$\begin{aligned} \bar{Q}_m(t) &= 2\alpha \frac{p^{0,0}(t)}{\text{St}_h \epsilon^2} N_m^2 e^{-\frac{\beta_m^2}{\text{Re}_h \text{St}_h} \int_0^t \frac{ds}{p^{0,0}(s)}} \times \\ &\quad \int_0^t \Delta p(s) e^{\frac{\beta_m^2}{\text{Re}_h \text{St}_h} \int_0^s \frac{dz}{p^{0,0}(z)}} (p^{0,0}(s))^{-2} ds \end{aligned} \quad (16)$$

where

$$N_m = \frac{(-1)^m}{(m+1/2)\pi},$$

and $\beta_m = (2m+1)\pi$, $m = 0, 1, \dots$.

The steady solution in this case reduces to the classical Poiseuille solution

$$\bar{Q}^{steady} = \sum_m \frac{2\alpha Re_h}{\beta_m^2 \epsilon^2} \frac{\Delta p}{(m + \frac{1}{2})^2 \pi^2} = \frac{\alpha Re_h \Delta p}{12 \epsilon^2} \quad (17)$$

Obtaining inlet and outlet volume flow rates

We use Eqn.(6) to obtain expressions for Q_{in} and Q_{out} as a function of $\bar{Q}(t)$. When we integrate Eqn.(6) in y , we get

$$\frac{1}{p^{0,0}} \frac{dp^{0,0}}{dt} + \frac{\partial Q}{\partial x} = 0 \quad (18)$$

Upon integrating Eqn.(18) in x from inlet to outlet, we obtain

$$Q_{out} - Q_{in} = -\frac{1}{p^{0,0}} \frac{dp^{0,0}}{dt} \quad (19)$$

whereas if we integrate Eqn.(18) from inlet to x , we get

$$Q(x,t) - Q_{in} = -\frac{1}{p^{0,0}} \frac{dp^{0,0}}{dt} x \quad (20)$$

And finally integrating Eqn.(20) from inlet to outlet, we get the desired

$$Q_{in}(t) = \bar{Q}(t) + \frac{1}{2p^{0,0}(t)} \frac{dp^{0,0}}{dt} \quad (21)$$

and so

$$Q_{out}(t) = \bar{Q}(t) - \frac{1}{2p^{0,0}(t)} \frac{dp^{0,0}}{dt} \quad (22)$$

MODEL VALIDATION AND DISCUSSION

The model shown in Fig. 2 enables system level analysis of channel flow using the conventional techniques employed in circuit analysis. Direct Current (DC) analysis is steady state analysis in which the transient term is set to zero and all the inductors in Figure 2 are short circuited. Transient analysis can also be performed by integrating the ODEs in time using circuit simulation tools such as Spice.

Accuracy and performance of the model presented in the previous section are addressed in terms of steady and transient isothermal flow of air in the channel depicted in Fig. 3 for different operating conditions. The channel length and height are

respectively $2000 \mu\text{m}$ and $20 \mu\text{m}$. The channel is preceded by an entry section in order for the flow to reach fully developed conditions at the inlet to the main section. The boundary condition at the inlet to the entry section is prescribed in terms of time varying inlet mass flux ($\rho_i V_i$), assumed to be spatially uniform. At the outlet of the main section, atmospheric pressure is prescribed.

In the first set of simulations, $\rho_i V_i = (\rho_i V_i)_{max}(1 - \exp(-t/\tau))$, where $\tau = L/V_{i,max}$. The values of Reynold number corresponding to $V_{i,max}$ are respectively 13.45, 29.3, 68.3 and 117. For each case, the pressure history at the inlet to the main section ($p_i(t)$), shown in Fig. 4, is recorded from the CFD (Fluent) transient simulation. $p_i(t)$ is then used as input to the model in order to compute the corresponding time varying mean volume flow rate. The volume flow rate computed from the reduced order model is in good agreement with that obtained from CFD (Fluent), as shown in Figure 5.

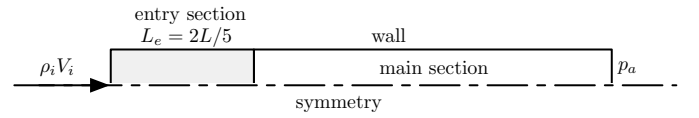


Figure 3. Schematic.

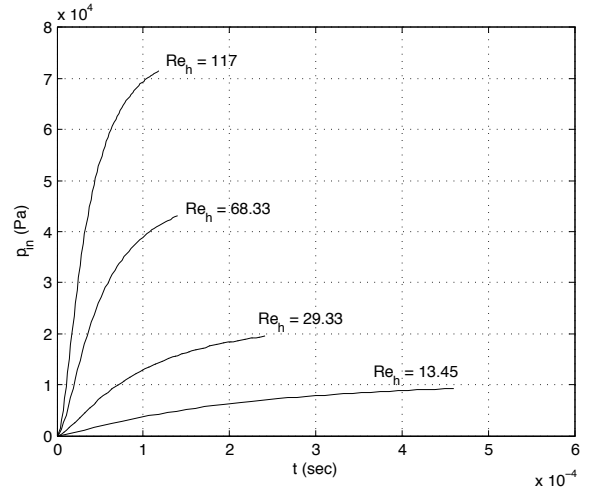


Figure 4. Pressure at inlet to main section for Set1.

In the second set, the mass flux at the entry section inlet is given a prescribed harmonic time variation according to $\rho_i V_i = (\rho_i V_i)_{max} \sin(2\pi f t)$, where the frequency is chosen according to $f^{-1} = L/V_{i,max}$ so that $St = 1$. For this case, another

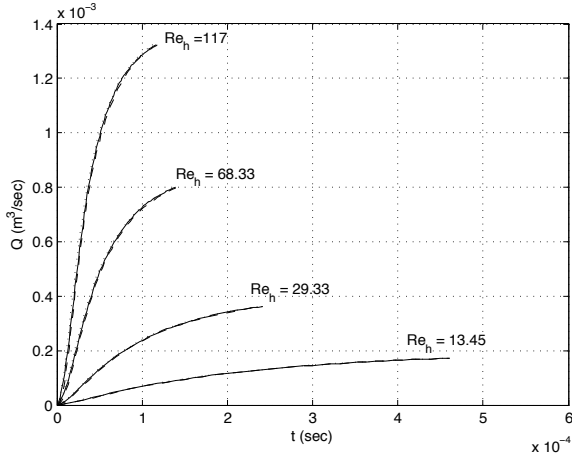


Figure 5. Average volume flow rate predicted by the model (dashed line) compared with FLUENT (solid line).

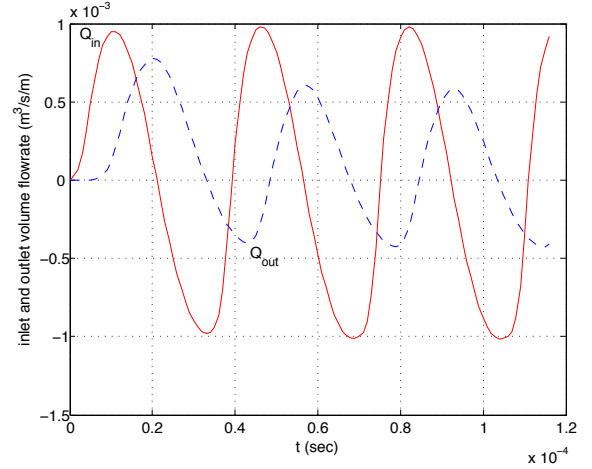


Figure 7. Volume flow rate at inlet and outlet of main section versus time.

section is added to the outlet of the main section with length of $2L/5$. The pressure at inlet and outlet of main section, as obtained from CFD (Fluent), is shown in Fig. 6. The volume flow rates at the inlet and outlet of the main section is presented versus time in Fig. 7 for the case of $Ma = 0.162$ and $Re = 117$. Time

ferred by the agreement with the CFD solution. This is expected because the model is developed for low Mach number flows. It should be noted that the simulation time using the model is a very small fraction of that of the CFD simulation tool. For example, the ratio of the model simulation time to the CFD simulation time for the case $Ma = 0.054$, $Re = 29.3$ is 1/280.

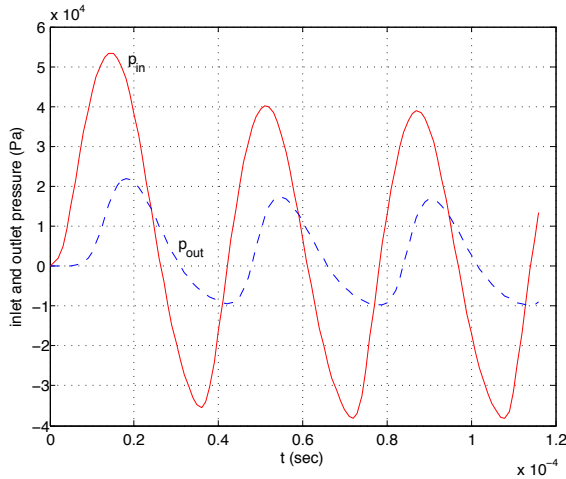


Figure 6. Pressure at inlet and outlet of main section versus time.

evolution of the mean volume flow rate is plotted in Fig. 8 where the numerical (CFD) solution is compared to that predicted by the proposed model for the three cases (i) $Ma = 0.013$, $Re = 6.4$, (ii) $Ma = 0.054$, $Re = 29.3$ and (iii) $Ma = 0.162$, $Re = 117$. It can be observed from Fig. 8 that the model yields more accurate results at smaller values of Mach and Reynolds number, as in-

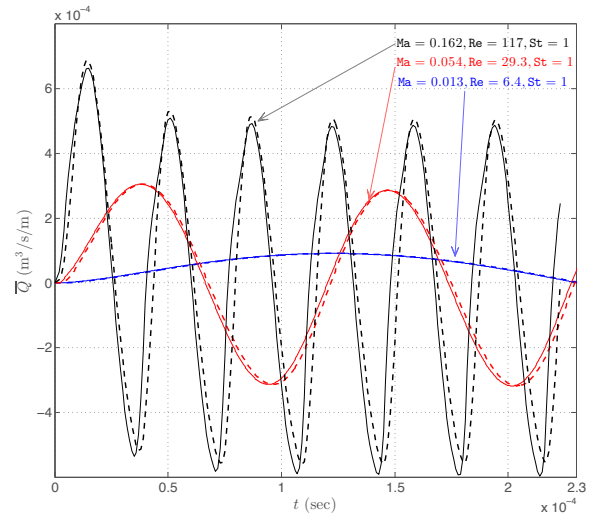


Figure 8. Comparison between time evolution of mean volume flow rate between model (dashed line) and CFD (solid line).

CONCLUSION

Reduced order models for unsteady low Mach number isothermal flow of an ideal gas in a two-dimensional parallel channel are presented for no-slip as well as first order slip boundary conditions. Each model consists essentially of a set of parallel branches, each consisting of a time varying resistance and a time varying inductor in series. The resistors model viscous dissipation and the inductors correspond to inertia due to the time varying term. The accuracy and validity of the proposed model are investigated in terms of dynamic response of the volume flow rate for sinusoidal as well as step function pressure history. The proposed model offers a tradeoff between speed and accuracy is possible by controlling the number of parallel branches included in the model.

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